

Countable Additivity is Not First-Order Axiomatizable

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Abstract

In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ -additive measure on the generated σ -algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof is a single ultraproduct argument: Dirac charges on the finite-cofinite algebra are individually σ -additive, but their ultraproduct is the finite-cofinite charge, which fails σ -additivity — contradicting Łoś’s theorem if any first-order axiomatization existed. The same obstruction shows that collective exhaustion — the necessary and sufficient condition for σ -additive extension in a directed system of Boolean algebras — cannot be derived from any first-order structural condition.

A property of structures is first-order axiomatizable in a language L if there exists a set of L -sentences whose models are exactly the structures satisfying it. Łoś’s theorem gives the key diagnostic: if a property fails in an ultraproduct of structures that individually satisfy it, no first-order axiomatization exists.

Countable additivity distinguishes σ -additive measures from finitely additive charges. That it cannot be expressed by a single first-order sentence is clear: the condition quantifies over countable sequences, unavailable in first-order logic. The probability logic community has treated the non-first-order character of σ -additivity as background knowledge: Hoover [4] requires an infinitary continuity rule to axiomatize σ -additive probability logic, Keisler [5] notes that σ -additivity does not fit finitary Henkin constructions, and Fagin, Halpern, and Megiddo [3] remark without proof that countable additivity “cannot be expressed by a formula” in the natural language of Boolean algebras with rational charge comparisons. The sharper question — whether countable additivity admits a first-order *axiomatization* by any set of sentences in that language, not merely a single formula — does not appear to have been settled by an explicit proof in the literature.

We settle it negatively.

Let $\mathcal{L}_{\text{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0, 1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\text{BA},\mu}$ is a Boolean algebra B together with a $[0, 1]$ -valued finitely additive probability μ , encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0, 1]$ avoids introducing a second numeric sort.

Definition 1 (First-order theory of finitely additive probabilities). Let T_{fa} be the $\mathcal{L}_{\text{BA},\mu}$ -theory axiomatizing:

- B is a Boolean algebra;
- normalization: $\mu(0) = 0$ and $\mu(1) = 1$;

(iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;

(iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \Rightarrow M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\text{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n \in \mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\text{BA},\mu}$.

Proposition 2 (σ -additive extension is not first-order axiomatizable). *There is no first-order $\mathcal{L}_{\text{BA},\mu}$ -theory $T \supseteq T_{\text{fa}}$ such that, for every $(B, \mu) \models T_{\text{fa}}$,*

$$(B, \mu) \models T \iff \mu \text{ extends to a } \sigma\text{-additive measure on } \sigma(B).$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let E be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \rightarrow \mathbb{Q}$, and let δ_n be the Dirac charge at $q(n)$. Each (E, δ_n) is a model of T_{fa} in which δ_n is σ -additive. Suppose for contradiction that a theory T as described exists; then each $(E, \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [6], the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ also models T . The charge induced on the diagonal copy of E is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n) \in A}.$$

If A is finite, $q(n) \in A$ for only finitely many n , so $\ell(A) = 0$. If A is cofinite, $q(n) \in A$ for cofinitely many n , so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus ℓ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q} \setminus \{q(0), \dots, q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \dots$ and $\bigcap_k A_k = \emptyset$, yet $\ell(A_k) = 1$ for all k . Hence the charge on the diagonal copy of E fails σ -additivity and does not extend to a σ -additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory T characterizing σ -additive extension, contradicting Łoś's theorem applied to the assumption that each $(E, \delta_n) \models T$. $\square$

Remark 3. The ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ is an $\mathcal{L}_{\text{BA},\mu}$ -structure in its own right; ℓ is the charge induced on the diagonal copy of E inside it. Identifying the ultraproduct with (E, ℓ) is harmless for the contradiction, since the failure of σ -additivity is witnessed on the diagonal copy.

Let $(\mathcal{E}_i, \ell_i)_{i \in I}$ be a directed system of Boolean charge spaces: each $\mathcal{E}_i$ is a Boolean algebra and $\ell_i : \mathcal{E}_i \rightarrow [0, 1]$ is a normalized finitely additive charge, with compatible connecting maps. *Collective exhaustion* (CE) is the following condition on the induced charge ℓ on the direct limit algebra $\mathcal{E} = \varinjlim \mathcal{E}_i$: whenever $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$, we have $\ell(E_n) \rightarrow 0$. CE is a valuation-layer condition: it constrains how charges behave across the directed system, not the Boolean-algebraic index structure. It is the necessary and sufficient condition under which a compatible finitely additive family extends to a σ -additive measure on the observable σ -algebra generated by the system.

Corollary 4 (CE is not first-order). *In the first-order language $\mathcal{L}_{\text{BA},\mu}$, no first-order theory characterizes those models satisfying CE. Equivalently, no accumulation of first-order conditions on the Boolean-algebraic structure can imply collective exhaustion.*

The finite-cofinite charge is the canonical witness: it satisfies every first-order structural condition yet assigns persistent unit mass to the decreasing sequence $A_k = \mathbb{Q} \setminus \{q(0), \dots, q(k)\}$, whose intersection is empty. CE fails precisely because the mass is not exhausted.

Foundational disputes about probability are not terminological. They concern which extra structure licenses the passage from finite coherence to probability, and Corollary 4 shows that some such structure is logically unavoidable: no first-order condition can close the gap. For a full development of CE as the admissibility condition for σ -additive extension in a directed observational system, see [7].

Proposition 2 shows that *one* purely finitely additive charge arises as an ultraproduct of σ -additive ones — that single witness suffices for the non-axiomatizability conclusion. A stronger question concerns the scope of this phenomenon:

Which purely finitely additive charges on a Boolean algebra arise as pointwise ultra-limits of σ -additive probabilities on the same algebra?

On a σ -algebra the answer is: none. If (μ_n) is a sequence of σ -additive probabilities on a σ -algebra Σ and $\mathcal{U}$ is any ultrafilter on $\mathbb{N}$, then $\ell(A) = \lim_{\mathcal{U}} \mu_n(A)$ is itself σ -additive. The reason is that for disjoint (A_k) with union A , the identity $\mu_n(A) = \sum_k \mu_n(A_k)$ holds exactly for each n , so the limit and sum interchange without difficulty; see Dunford–Schwartz [2], IV.9.8. Consequently the question is entirely about Boolean algebras that are not σ -complete.

On a general Boolean algebra the situation is open. The Yosida–Hewitt decomposition [8] gives $\text{ba} = \text{ca} \oplus \text{pfa}$ (orthogonal sum of σ -additive and purely finitely additive parts) but says nothing about approximation. For ultrafilter-generated $\{0, 1\}$ -valued charges the present proof already gives a positive answer: each such charge is the ultralimit of the Dirac masses on the finite-cofinite algebra. Whether every purely finitely additive charge on a Boolean algebra admits such a representation — and in particular whether the answer depends on set-theoretic hypotheses about ultrafilters — is left open. Duanmu and Weiss [1] show that on totally bounded separable metric spaces every finitely additive Borel probability is a weak limit (against bounded uniformly continuous test functions) of σ -additive probabilities, but remark that setwise ultralimit approximation fails for ultrafilter charges on discrete sets (Example 5.14), confirming that the pointwise ultralimit question is genuinely delicate.

References

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